Tunnel Monitoring System: An Automated LiDAR-Based Point Cloud Processing Pipeline for Structural Health Monitoring of Underground Tunnels

Keywords: LiDAR point cloud; tunnel structural health monitoring; automatic denoising; Frenet frame; M3C2 change detection; IFC4X3; retrieval-augmented generation

Abstract

Automated structural health monitoring (SHM) of underground tunnels using terrestrial LiDAR remains challenging due to the presence of non-structural clutter — cables, lighting fixtures, and personnel — that contaminates raw point clouds and causes systematic errors in geometric analysis. Conventional workflows require extensive manual preprocessing, limiting scalability to large tunnel networks. This paper presents the SSL Smart Tunnel Monitoring System, an end-to-end Python-based pipeline that automates the full analysis chain from raw LiDAR ingestion to engineering-grade report generation. The proposed system introduces three principal contributions: (1) a three-stage cascaded auto-denoising algorithm combining morphological classification, radial statistical filtering, and wall-cable protrusion detection that eliminates non-structural clutter without labelled training data; (2) a Frenet-frame-based cross-section extraction method that guarantees geometric orthogonality to the tunnel axis, removing an apparent ovality error of up to 15% present in axis-aligned slicing approaches; and (3) a local Retrieval-Augmented Generation (RAG) assistant powered by an on-device large language model (LLM) that translates structured measurement data into engineer-readable safety assessments without transmitting data to external servers. Experimental validation on real tunnel scan datasets demonstrates per-section deformation metric extraction — crown settlement, lateral convergence, ovality, and eccentricity — in full compliance with Korean Railway Safety Standards (KR C-08080) and Korean Design Standards for Tunnels (KDS 27 25 00). The pipeline produces IFC4X3-compatible Building Information Models and professional PDF reports suitable for direct handover to infrastructure owners.

1. Introduction

Underground tunnels constitute a critical node in modern transportation networks, yet they are also among the most vulnerable assets in aging infrastructure inventories. The catastrophic collapse of the Frejus road tunnel (France–Italy, 1999) and the Gleinalm Tunnel blowout (Austria, 2001) exposed the consequences of undetected lining deterioration, prompting the European Union to mandate periodic geometric inspections for all rail and road tunnels exceeding 500 m [1,2]. In South Korea, the national railway network operates more than 700 tunnels with a total length exceeding 850 km; KR C-08080 and KDS 27 25 00 require geometric surveys at intervals of not less than six months for tunnels in deformation-sensitive ground conditions [3,4]. The economic consequences of tunnel failures are equally severe: a single unplanned closure of a metropolitan rail tunnel results in a direct loss estimated at USD

1–3 million per day in Seoul's urban network [5]. These safety and economic imperatives make automated, high-accuracy structural health monitoring (SHM) of underground tunnels a critical engineering priority.

Terrestrial LiDAR scanning has emerged as the dominant technology for tunnel SHM owing to its ability to capture full-section, sub-millimetre-resolution 3D point clouds in a single survey pass without disrupting train operations [6,7]. Prior work has demonstrated the effectiveness of LiDAR-based approaches for specific subtasks: Jung et al. [8] extracted ovality metrics from mobile laser scanning data using iterative circle fitting; Gikas [9] applied least-squares cylinder fitting for long-term convergence monitoring; Ye et al. [10] combined 3D semantic segmentation with point cloud processing to detect surface cracks at millimetre scale; and Attard et al. [11] benchmarked five commercial software packages on tunnel inspection accuracy. Concurrent advances in point cloud registration — notably Generalised ICP [12] and graph-based outlier rejection [13] — have improved multi-station co-registration to sub-millimetre RMSE under sparse-feature tunnel conditions. Change detection between survey epochs has been formalised through the M3C2 algorithm [14], which provides statistically principled Level-of-Detection thresholding that accounts for local point cloud roughness. More recently, Retrieval-Augmented Generation (RAG) architectures [15] have been applied to structural assessment by grounding large language model (LLM) outputs in retrieved engineering standards, substantially reducing hallucination rates in safety-critical contexts [16,17].

Despite these individual advances, three persistent gaps prevent their deployment as a unified, production-grade inspection pipeline. First, raw tunnel scans contain 5–30% non-structural points — cables, lighting fixtures, survey targets, and personnel — that corrupt downstream geometric estimators if not removed; existing statistical outlier removal methods [18] are designed for random Gaussian noise and fail against the structured, elongated geometry of wall-mounted cable runs. Second, cross-section extraction in curved tunnels requires slicing perpendicular to the local tunnel axis: axis-aligned (world-frame) sectioning introduces oblique cuts that systematically overestimate ovality by up to 15% in arcs with radius below 300 m, yet no open-source tool automatically applies axis-orthogonal Frenet-frame sectioning. Third, translating geometric metrics — crown settlement δ_v, lateral convergence δ_h, ovality ε, and eccentricity e — into prioritised engineering actions currently demands manual review by a qualified structural engineer for every report cycle, creating a bottleneck that limits monitoring frequency and scalability. No existing open-source system addresses all three gaps within a single, standards-compliant, end-to-end pipeline.

This paper presents the SSL Smart Tunnel Monitoring System, a fully automated, Python-based LiDAR processing pipeline that closes all three gaps simultaneously. The system ingests raw multi-station scans in standard formats (LAS, LAZ, PLY, TXT), applies a novel cascaded denoising algorithm, performs robust multi-scan registration, extracts geometrically correct cross-sections via Frenet frames, computes all KR C-08080–specified deformation metrics, and produces engineering-grade outputs — CSV/Excel summaries, professional PDF reports, and IFC4X3 Building Information Models — without any manual intervention. A local Retrieval-Augmented Generation assistant, powered by an on-device LLM, generates engineer-readable safety assessments from structured measurement data without transmitting any data to external servers.

The principal contributions of this paper are as follows:

1. **A three-stage cascaded auto-denoising algorithm** — combining morphological PCA-based classification, radial median-absolute-deviation (MAD) filtering, and cylindrical-grid wall-cable detection — that eliminates non-structural clutter from raw tunnel scans without any labelled training data or manual parameter tuning.

2. **A Frenet-frame cross-section extraction method** that guarantees geometric orthogonality to the instantaneous tunnel axis, removing the systematic ovality bias of up to 15% introduced by world-frame slicing in curved tunnel segments.
3. **A local RAG-LLM engineering assistant** that retrieves relevant Korean railway safety standard excerpts (KR C-08080, KDS 27 25 00) and generates prioritised maintenance recommendations entirely on-device, with no dependency on external API services.
4. **An end-to-end open-source pipeline** producing IFC4X3-compatible BIM models, professional PDF inspection reports, and structured CSV/Excel output from a single LiDAR scan, validated against real tunnel datasets and compliant with Korean railway safety standards.

The remainder of this paper is organised as follows. Section 2 surveys related work across the four constituent domains. Section 3 describes the overall system architecture and module interfaces. Sections 4 through 8 present each processing stage in detail, including the denoising cascade (Section 4), multi-scan registration (Section 5), Frenet-frame geometric analysis (Section 6), parameter extraction (Section 7), and multi-epoch change detection (Section 8). Section 9 covers the output generation modules. Section 10 reports experimental validation on real tunnel scan datasets. Section 11 concludes with directions for future work.

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